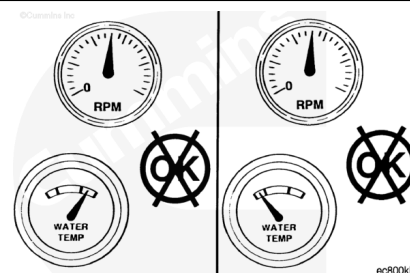


Trainings ▾

Initial Check

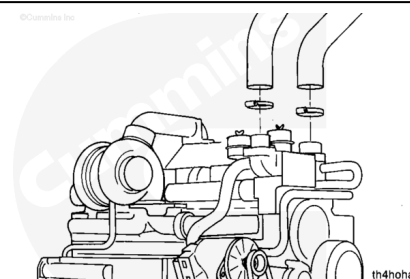
The engine thermostat and thermostat seal **must** operate properly in order for the engine to operate in the most efficient heat range. Overheating or overcooling will shorten engine life.

Note : With a low temperature aftercooling thermostat, the intake air temperature is monitored by the electronic system and will generate a fault code when the intake air temperature exceeds the maximum temperature limits.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [122°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Complete this test with the engine coolant temperature below 50°C [122°F]. Heated coolant spray or steam can cause serious personal injury.

⚠ CAUTION ⚠

Use caution when draining coolant that the coolant is not spilled or drained into the bilge area. Do not pump coolant overboard. If the coolant is not reused, it must be disposed of in accordance with local environmental regulations

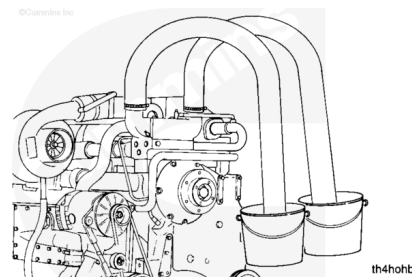
Note : If the engine being tested has low temperature aftercooling, leave the low temperature aftercooling radiator hoses connected to the thermostat housing.

Remove the upper engine radiator hoses from the thermostat housing.

Install two hoses of the same size, long enough to reach a remote dry container, on the thermostat housing outlets.

Install and tighten hose clamps on the housing outlets.

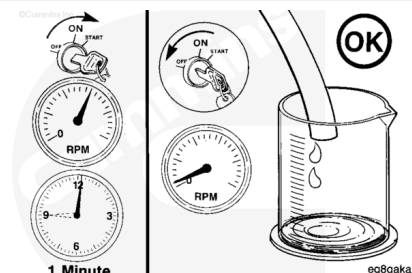
Put the end of the hoses in two dry containers.



Operate the engine at rated rpm for one minute.

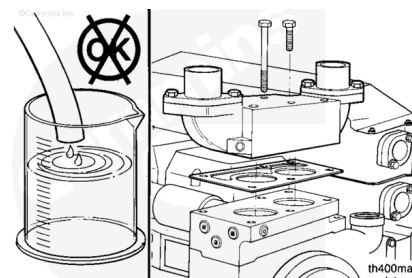
Shut the engine OFF, and measure the amount of coolant collected in each container.

The amount of coolant collected **must not** be more than 100 cc [3.3 fl oz].

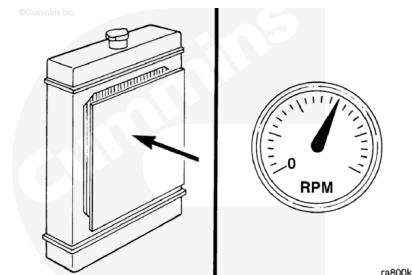


If more than 100 cc [3.3 fl oz] of coolant is collected, the thermostat or the thermostat seal is leaking.

Remove the thermostat and test operation. Refer to Procedure 008-013 (</qs3/pubsys2/xml/en/procedures/20/20-008-013-tr.html>).

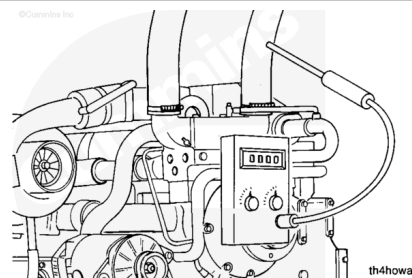


Restrict the radiator air flow. Operate the engine until the coolant temperature rises to 90 to 93°C [195 to 200°F].

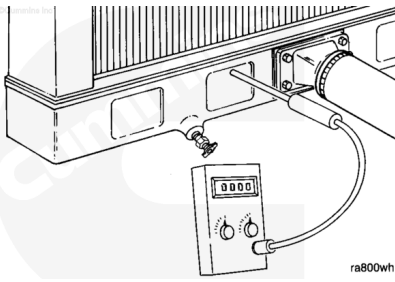


Record the temperature of the coolant outlet hoses. An increase in temperature indicates the thermostats have started to open.

Use a contact pyrometer (shown) or install a temperature gauge in the desired locations prior to operating the engine.



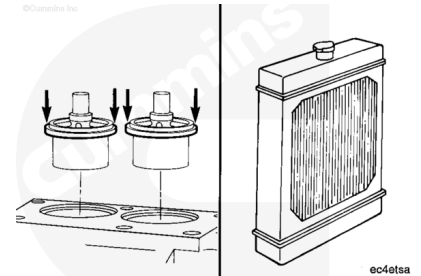
Record the temperature of the radiator bottom tank or coolant outlet tube.



If the difference in temperature is more than 8°C [15°F], either the thermostats are **not** fully open or the radiator core is plugged.

If **not** within specification; replace the thermostats. Refer to Procedure 008-013 (/qs3/pubsys2/xml/en/procedures/20/20-008-013-tr.html).

Check the radiator. Refer to Procedure 008-042 (/qs3/pubsys2/xml/en/procedures/20/20-008-042.html).



Preparatory Steps

⚠ WARNING ⚠

Allow the engine coolant temperature to cool to below 50°C [122°F] before draining. Heated coolant can cause personal injury.

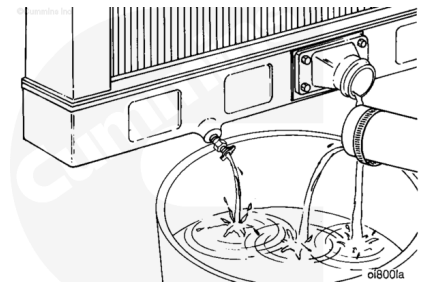
⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [122°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ CAUTION ⚠



Use caution when draining coolant that the coolant is not spilled or drained into the bilge area. Do not pump coolant overboard. If the coolant is not reused, it must be disposed of in accordance with local environmental regulations

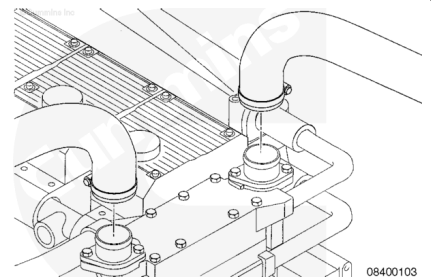
Drain the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>).

Remove

Industrial Applications

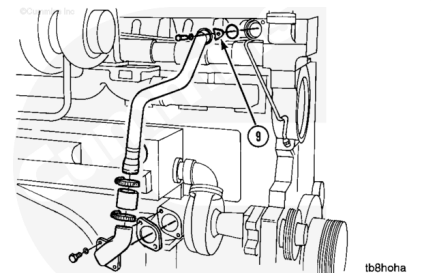
Remove both upper radiator hoses from the thermostat housing.

Remove the vent lines from the top of the thermostat housing.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



Note : If the engine is equipped with an air compressor, remove the air compressor coolant return tube.

Conventional Aftercooling

Remove bypass tube clip (9).

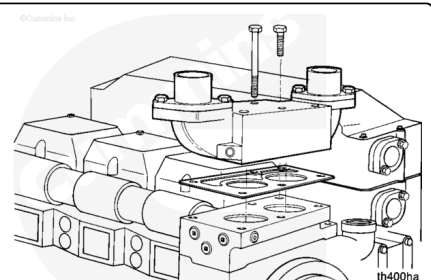
Loosen both hose clamps.

Remove the bypass tube.

Remove the eight long capscrews.

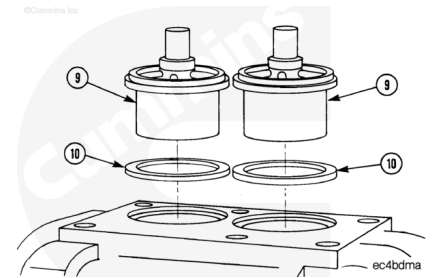
Remove the thermostat housing.

Remove and discard the gasket.



Remove the thermostats (9).

Remove the seals (10). Refer to Procedure 008-016
(/qs3/pubsys2/xml/en/procedures/20/20-008-016-tr.html).

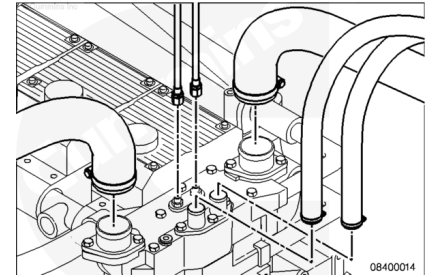


The following is for Low Temperature Aftercooled (LTA) engines:

Remove both of the upper engine radiator hoses from the thermostat housing.

Remove both of the upper low temperature aftercooling radiator hoses from the thermostat housing.

Remove both of the vent lines from the top of the thermostat housing.

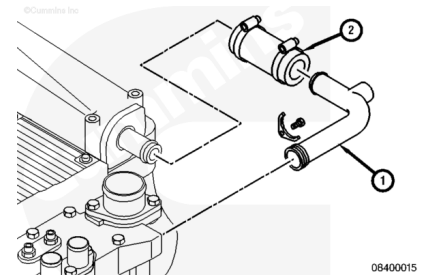


⚠ WARNING ⚠

Allow the engine coolant temperature to cool to below 50°C [122°F] before draining. Heated coolant can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



The following is for Low Temperature Aftercooled (LTA) engines:

Remove the aftercooler return tube clip.

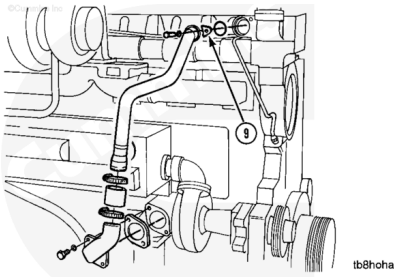
Loosen both hose clamps (2).

Remove the aftercooler coolant return tube (1).

Remove the bypass tube clip (9).

Loosen both hose clamps.

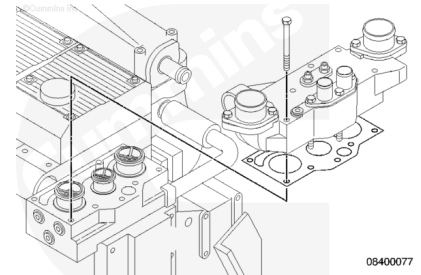
Remove the bypass tube.



Remove the nine long cap screws.

Remove the thermostat housing.

Remove and discard the gasket.



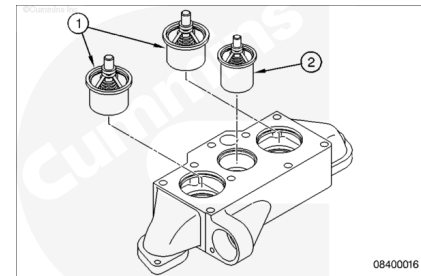
Remove both of the engine thermostats (1).

Remove the low temperature aftercooling thermostat (2).

Remove the seals.

Refer to Procedure 008-016

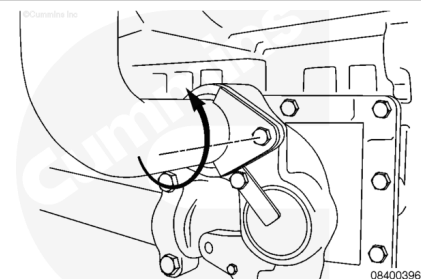
(</qs3/pubsys2/xml/en/procedures/20/20-008-016-tr.html>).



with Mechanically Actuated Injector

Marine Applications

Loosen the retaining clip on the lower end of the Low Temperature Aftercooling (LTA) supply pipe.



Remove the three captive washer cap screws from the water transfer connection.

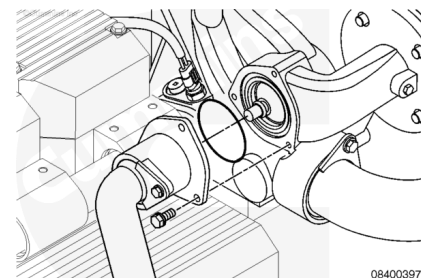
Pull the water transfer connection away from the thermostat housing.

The pipe will rotate around the lower end.

Discard the o-ring.

Remove the thermostat.

Remove the thermostat seal.



Refer to Procedure 008-016

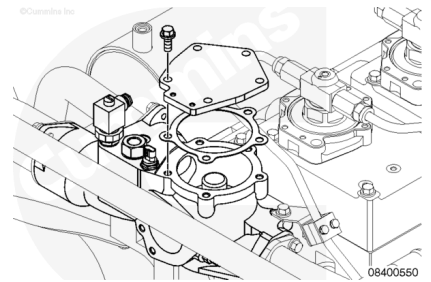
(/qs3/pubsys2/xml/en/procedures/20/20-008-016-tr.html).

with Electronically Actuated Injector

Marine Applications

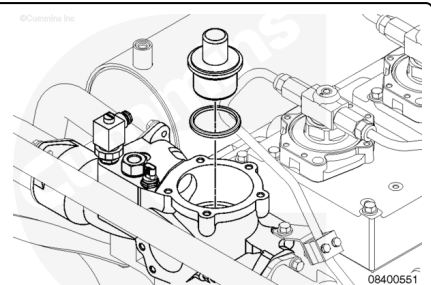
Remove the five capscrews from the thermostat cover.

Remove the thermostat cover and discard the gasket.



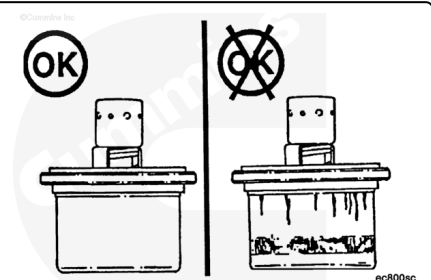
Remove the thermostat from the thermostat housing.

Remove the thermostat seal. Refer to Procedure 008-016
(/qs3/pubsys2/xml/en/procedures/20/20-008-016-tr.html).



Inspect for Reuse

Inspect the thermostat for damage.



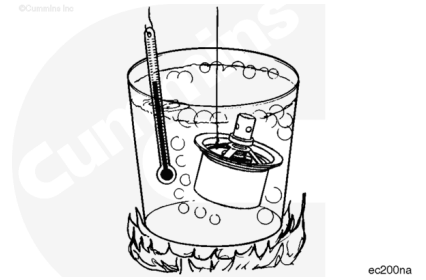
Test

Industrial Applications

Suspend the thermostat and a 100°C [212°F] thermometer in a container of water.

Note : Do not allow the thermostat or thermometer to touch the container.

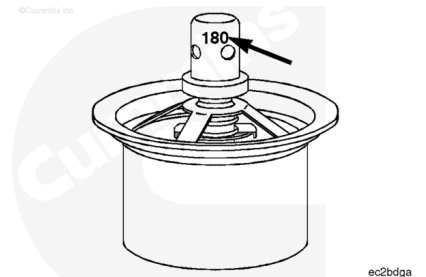
Heat the water and check the thermostat as follows.



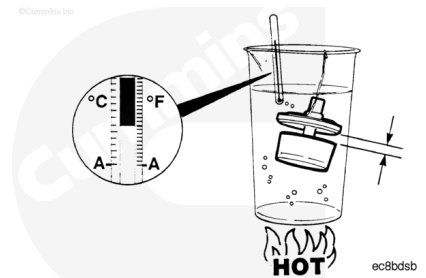
The nominal operating temperature is stamped on the thermostat.

- Thermostat **must** begin to open within 2°C [3°F] of nominal temperature.
- Thermostat **must** be fully open at 12°C [22°F] above nominal temperature.

The fully open distance between the thermostat flange and housing is 11.05 mm [0.43 in].



Remove the container from the heat. Check to see if the thermostat returns to the closed position.

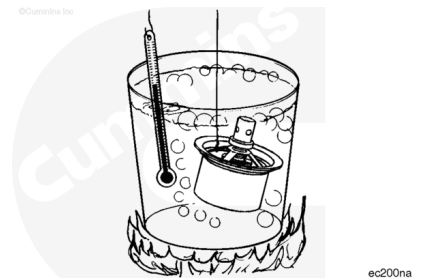


Note : Do not allow the thermostat or thermometer to touch the container.

The following is for Low Temperature Aftercooled (LTA) engines:

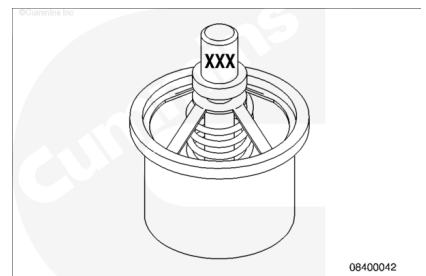
Suspend the thermostat and a 100°C [212°F] thermometer in a container of water.

Heat the water and check the thermostat as follows.



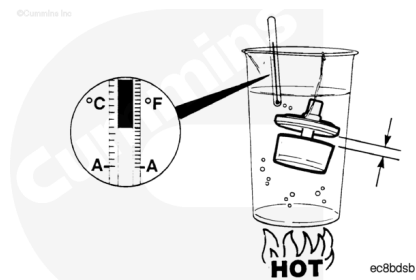
The nominal operating temperature is stamped on the thermostat.

- Thermostat **must** begin to open within 2°C [3°F] of nominal temperature.
- Thermostat **must** be fully open at 8°C [15°F] above nominal temperature.



The fully open distance between the thermostat flange and housing is 7.8 mm [0.3 in].

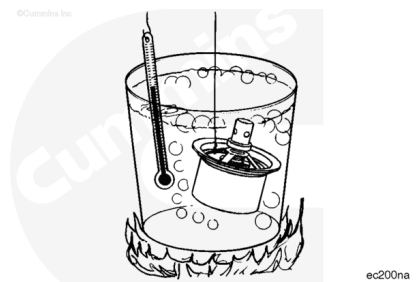
Remove the container from the heat. Check to see if the thermostat returns to the closed position.



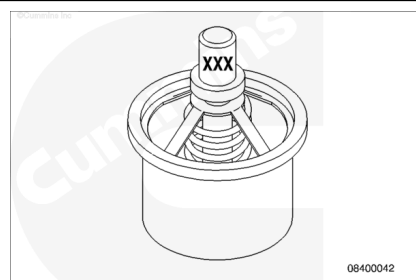
Marine Applications

Suspend the single loop Low Temperature Aftercooling (LTA) thermostat and a 100°C [212°F] thermometer in a container of water.

Do **not** allow the thermostat or thermometer to touch the container.



Heat the water and check the thermostat as follows:
Nominal temperature is 66°C [150°F].



Measurements

	celsius	fahrenheit
Thermostat begins to open:	63 to 67	145 to 152

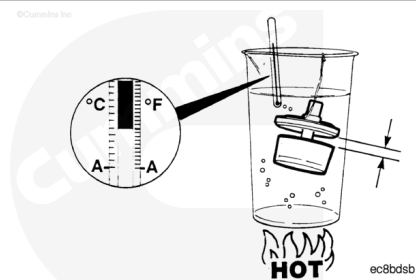
Measurements

	celsius	fahrenheit
Thermostat fully open:	79	175

The fully open distance is 9.5 mm [0.375 in].

Remove the container from heat.

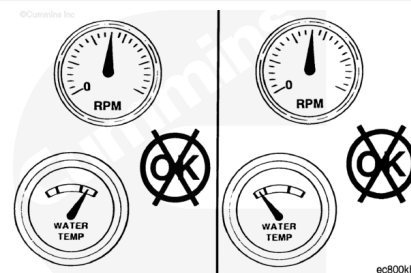
Make sure the thermostat returns to the closed position.



Leak Test

The engine thermostat and thermostat seal **must** operate properly in order for the engine to operate in the most efficient heat range. Overheating or overcooling will shorten engine life.

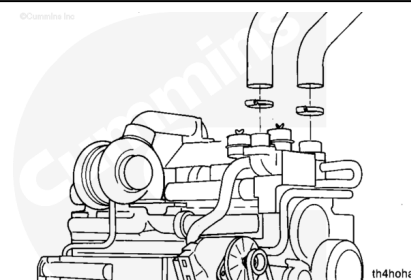
With a low temperature aftercooling thermostat, the intake air temperature is monitored by the electronic system and will generate a fault code when the intake air temperature exceeds the maximum temperature limits.



Engine Coolant Temperature

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the temperature is below 50°C [122°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ CAUTION ⚠

Use caution when draining coolant that the coolant is not spilled or drained into the bilge area. Do not pump coolant overboard. If the coolant is not reused, it must be disposed of in accordance with local environmental regulations

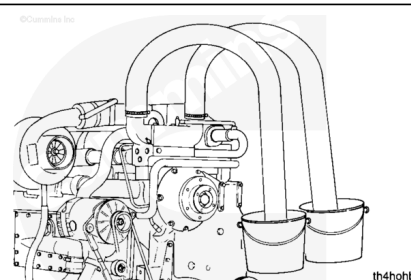
Note : If the engine being tested has low temperature aftercooling, leave the low temperature aftercooling radiator hoses connected to the thermostat housing.

Remove the upper engine radiator hoses from the thermostat housing.

Install two hoses of the same size, long enough to reach a remote dry container, on the thermostat housing outlets.

Install and tighten hose clamps on the housing outlets.

Put the end of the hoses in the two dry containers.

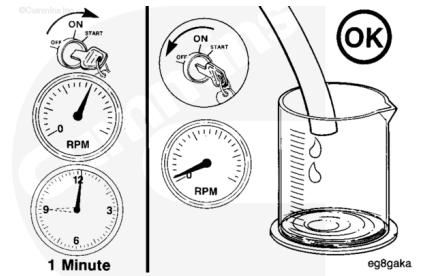


Operate the engine at rated rpm for 1 minute.

Shut the engine off.

Measure the amount of coolant collected in each container.

The amount of coolant collected **must not** be more than 100 cc [3.3 fl oz].

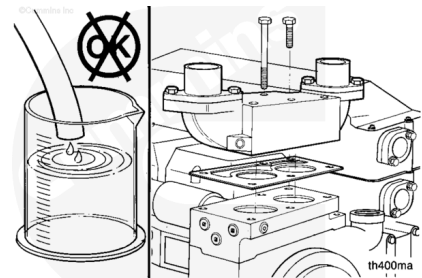


If more than 100 cc [3.3 fl oz] of coolant is collected, the thermostat or the thermostat seal is leaking.



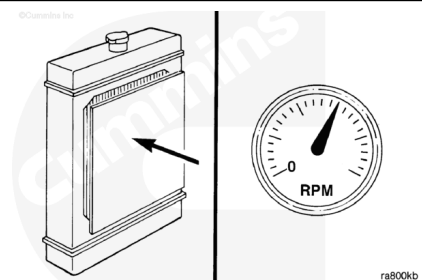
Remove the thermostats and replace the seals. Refer to Procedure 008-015 (</qs3/pubsys2/xml/en/procedures/20/20-008-015-tr.html>).

Test the thermostats for proper operation.



High Coolant Temperature

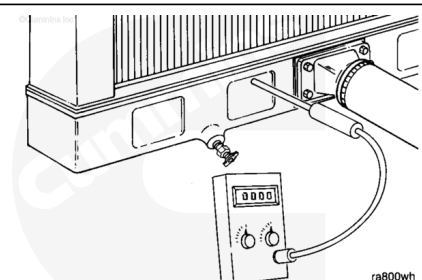
Restrict the radiator airflow. Operate the engine until the coolant temperature reaches 90 to 93°C [195 to 200°F].



Use a contact pyrometer (shown), or install a temperature gauge in the desired locations.

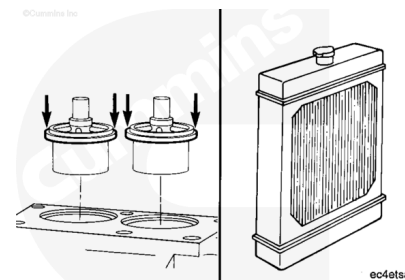
Record the temperature of the coolant outlet hoses. An increase in temperature indicates the thermostats have started to open.

Record the temperature of the radiator bottom tank or coolant out tube.



If the difference in temperature is more than 8°C [15°F], either the thermostats are **not** fully open or the radiator core is plugged.

Check the radiator. Refer to Procedure 008-042 (</qs3/pubsys2/xml/en/procedures/20/20-008-042.html>). If the radiator is in good working order, replace the thermostats.



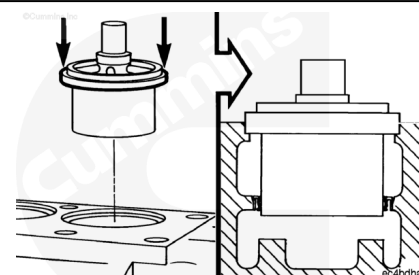
Install

Industrial Applications

The following is for conventional aftercooling engines:

Install the thermostat seals. Refer to Procedure 008-016 (</qs3/pubsys2/xml/en/procedures/20/20-008-016-tr.html>).

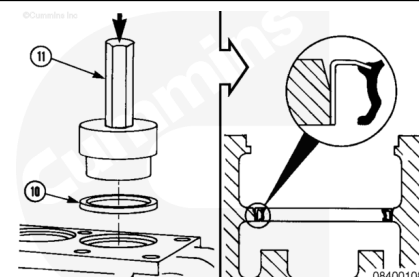
Install the thermostat by pushing on the outer rim.



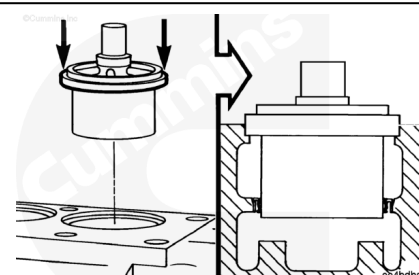
⚠ CAUTION ⚠

The seal must be installed with the part number pointing up.

Use a mallet and thermostat seal driver, Part Number 3375411, or equivalent. Install the thermostat seal no more than 0.51 mm [0.020 in] below the top of the cast edge.



Install the thermostat by pushing on the outer rim.

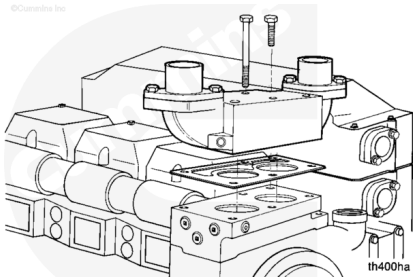


Install a new gasket.

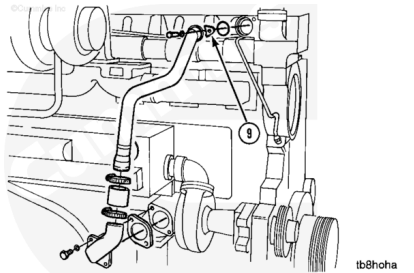
Install the thermostat housing and capscrews.

Tighten the capscrews.

Torque Value: 45 n•m [35 ft-lb]



Use vegetable oil to lubricate the o-ring on the bypass tube.
Install the bypass tube.
Install the retainer (9) and capscrew.
Tighten the capscrew and the hose clamps.



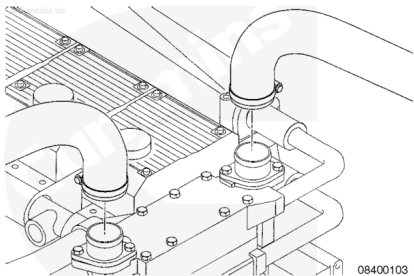
Capscrew	45 n.m	[35 ft-lb]
Hose Clamps	6 n.m	[50 in-lb]

Install the two upper radiator hoses.
Tighten the hoses clamps.

Torque Value: 6 n•m [50 in-lb]

Install the vent lines.

Note : If the engine is equipped with an air compressor, install the air compressor coolant return tube.

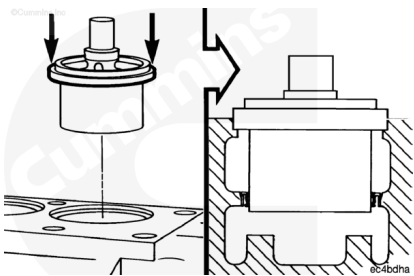


The following is for Low Temperature Aftercooling (LTA) engines:

Install the thermostat seals. Refer to Procedure 008-016 (/qs3/pubsys2/xml/en/procedures/20/20-008-016-tr.html).

Install two engine thermostats and one Low Temperature Aftercooling (LTA) thermostat.

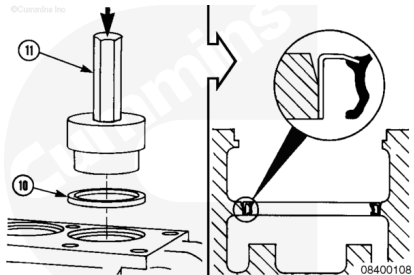
Install the thermostat by pushing on the outer rim.



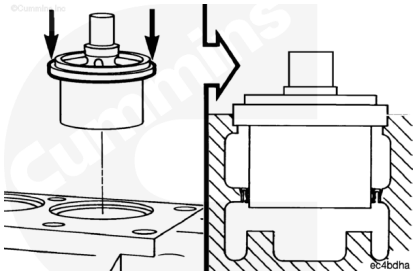
⚠ CAUTION ⚠

The seal must be installed with the part number pointing up.

Use a mallet and thermostat seal driver, Part Number 3375411, or equivalent. Install the thermostat seal no more than 0.51 mm [0.020 in] below the top of the cast edge.



Install the thermostat by pushing on the outer rim.

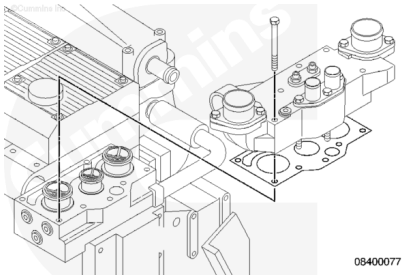


Install a new gasket.

Install the thermostat housing and capscrews.

Tighten the capscrews.

Torque Value: 45 n•m [35 ft-lb]

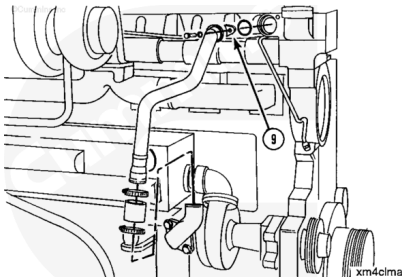


Use vegetable oil to lubricate the o-ring on the bypass tube.

Install the bypass tube.

Install the retainer (9) and capscrew.

Tighten the capscrew and the hose clamps.

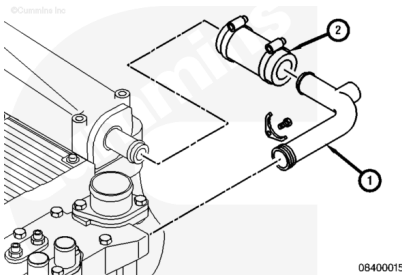


Capscrew	45 n.m	[35 ft-lb]
Hose Clamp	6 n.m	[50 in-lb]

Install the aftercooler water return tube (1), tube clip, and capscrew.

Install the connecting hose (2).

Tighten the hose clamps and capscrew.



Capscrew	20 n.m	[15 ft-lb]
Hose Clamp	6 n.m	[50 in-lb]

Install both vent lines to the top of the thermostat housing.

Install both upper low temperature aftercooling hoses.

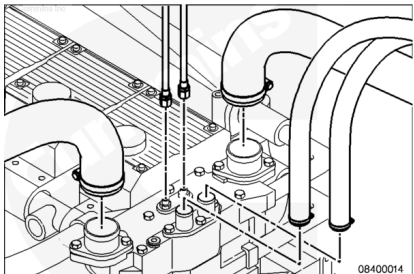
Tighten the hose clamps.

Torque Value: 6 n•m [50 in-lb]

Install both upper engine radiator hoses.

Tighten the hose clamps.

Torque Value: 6 n•m [50 in-lb]

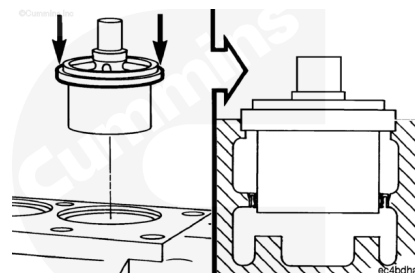


with Mechanically Actuated Injector

Marine Applications

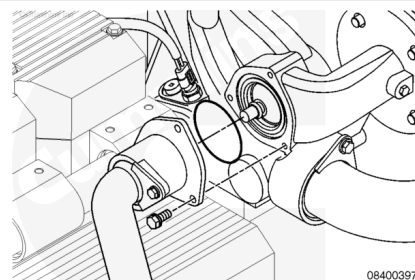
Install the thermostat seal. Refer to Procedure 008-016 (</qs3/pubsys2/xml/en/procedures/20/20-008-016-tr.html>).

Install the thermostat by pushing on the outer rim until it seats.



Use a new o-ring and fasten the thermostat housing supply pipe to the thermostat housing with three captive washer capscrews.

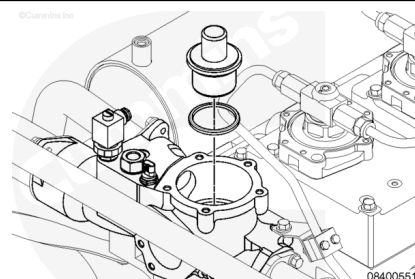
Torque Value: 40 n•m [30 ft-lb]



with Electronically Actuated Injector

Install the thermostat seal. Refer to Procedure 008-016 (</qs3/pubsys2/xml/en/procedures/20/20-008-016-tr.html>).

Install the thermostat into the thermostat housing.

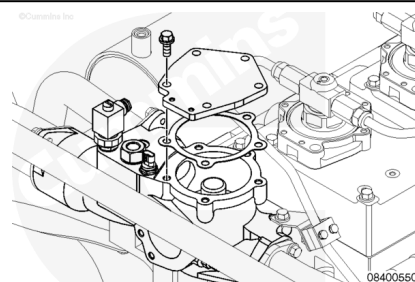


Install the thermostat cover and new gasket.

Install the five thermostat cover capscrews.

Tighten the capscrews.

Torque Value: 45 n•m [35 ft-lb]



Finishing Steps

⚠ CAUTION ⚠

Use caution when filling the cooling system so that the coolant is not spilled into the bilge area. Do not pump coolant overboard. If the coolant is spilled, it must be disposed of in accordance with local environmental regulations

- Fill the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>).
- Operate the engine to 70°C [160°F] coolant temperature and check for leaks.

